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(54) **STABILITY-ENHANCED PROTEASE
VARIANTS VI**

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(57) **ABSTRACT**

Proteases may include an amino acid sequence having at least 70% sequence identity with the amino acid sequence given in SEQ ID NO:1 over its entire length. The proteases may further have, in each case based on the numbering according to SEQ ID NO:1, amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271. The proteases may further include at least one further amino acid substitution at at least one of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224, and to the production and use thereof. Such proteases exhibit very good stability with good cleaning performance.

14 Claims, No Drawings

Specification includes a Sequence Listing.

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1

STABILITY-ENHANCED PROTEASE VARIANTS VI

REFERENCE TO A SEQUENCE LISTING SUBMITTED VIA EFS-WEB

The content of the ASCII text file of the sequence listing named "P84983US_Sequence_protocol", which is 6 kb in size was created on Mar. 3, 2020; the sequence listing is electronically submitted via EFS-Web herewith and is herein incorporated by reference in its entirety.

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a national stage entry according to 35 U.S.C. § 371 of PCT application No.: PCT/EP2021/054688 filed on Feb. 25, 2021; which claims priority to German patent application 10 2020 105 720.2, filed on Mar. 3, 2020; all of which are incorporated herein by reference in their entirety and for all purposes.

TECHNICAL FIELD

The disclosure relates to enzyme technology, such as proteases from *Bacillus pumilus*, the amino acid sequences of which have been altered to give them a better storage stability, in particular with regard to the use in washing and cleaning agents, and also relates to the nucleic acids coding for said proteases and to the production thereof. The disclosure further relates to the uses of these proteases and to methods in which they are used, as well as to agents containing them, in particular washing and cleaning agents.

BACKGROUND

Proteases are some of the technically most important enzymes. They are the longest established enzymes for washing and cleaning agents, and are contained in virtually all modern, effective washing and cleaning agents. They bring about the decomposition of protein-containing stains on the item to be cleaned. Of these, in turn, proteases of the subtilisin type (subtilases, subtilopeptidases, EC 3.4.21.62) are particularly important and are serine proteases due to the catalytically active amino acids. They act as non-specific endopeptidases and hydrolyze any acid amide bonds that are inside peptides or proteins. Their optimum pH is usually in the distinctly alkaline range. The article "Subtilases: Subtilisin-like Proteases" by R. Siezen, pages 75-95 in "Subtilisin enzymes," published by R. Bott and C. Betzel, New York, 1996, gives an overview of this family, for example. Subtilases are, naturally, formed from microorganisms. In particular, the subtilisins formed and secreted by *Bacillus* species are the most significant group of subtilases.

Examples of the subtilisin proteases preferably used in washing and cleaning agents are the subtilisins BPN[®] and Carlsberg, the protease PB92, the subtilisins 147 and 309, the protease from *Bacillus lentus*, in particular from *Bacillus lentus* DSM 5483, subtilisin DY and the enzymes thermitase, proteinase K and the proteases TW3 and TW7, which can be classified as subtilases but no longer as subtilisins in the narrower sense, and variants of said proteases having an amino acid sequence that has been altered with respect to the starting protease. Proteases are altered, selectively or randomly, by methods known from the prior art, and are thereby optimized for use in washing and cleaning agents, for example. These methods include point,

2

deletion or insertion mutagenesis, or fusion with other proteins or protein parts. Appropriately optimized variants are therefore known for the majority of proteases known from the prior art.

European patent application EP 2016175 A1 discloses, for example, a protease from *Bacillus pumilus* intended for washing and cleaning agents. In general, only selected proteases are suitable for use in liquid, surfactant-containing preparations in any case. Many proteases do not exhibit sufficient catalytic performance in such preparations. For the use of proteases in cleaning agents, therefore, a high catalytic activity under conditions as they are during a wash cycle and a high storage stability is particularly desirable.

Consequently, protease and surfactant-containing liquid formulations from the prior art are disadvantageous in that the proteases contained, under standard washing conditions (e.g. in a temperature range of from 20° C. to 40° C.), do not have satisfactory proteolytic activity or are not storage-stable and the formulations therefore do not exhibit optimal cleaning performance on protease-sensitive stains.

SUMMARY

Surprisingly, it has now been found that a protease from *Bacillus pumilus* or a sufficiently similar protease (based on the sequence identity) that has, based on the numbering according to SEQ ID NO:1, the amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271, and at least one further amino acid substitution, optionally selected from 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q and 224A/G, in particular from 6W, 89A, 131H, 166M, 189T, 211N and 224A, at at least one, preferably at least two, of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224, is improved in terms of storage stability compared with the wild-type form and/or reference mutants and is therefore particularly suitable for use in washing or cleaning agents.

In a first aspect, a protease may include an amino acid sequence which has at least 70% sequence identity with the amino acid sequence given in SEQ ID NO:1 over its entire length and has, in each case based on the numbering according to SEQ ID NO:1:

- (a) amino acid substitutions, in particular the amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E, at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; as well as
- (b) at least one further amino acid substitution, in particular selected from 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q and 224A/G, more preferably selected from 6W, 89A, 131H, 166M, 189T, 211N or 224A, at at least one, preferably at least two, of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224.

A method for producing a protease as defined above, may include the substitution of amino acids in a starting protease which has at least 70% sequence identity with the amino acid sequence given in SEQ ID NO:1 over its entire length (i) at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271 in SEQ ID NO:1, such that the protease comprises amino acid substitutions, in particular the amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E, at the positions, and (ii) has at least one further amino acid substitution, in particular selected from 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q and 224A/G, more preferably from 6W, 89A, 131H, 166M, 189T, 211N and 224A, at at least one, preferably at least two,

of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224 in SEQ NO:1. The protease that can be obtained by this method has at least 70% sequence identity with the amino acid sequence given in SEQ ID NO:1 over its entire length.

A protease within the meaning of the present patent application therefore comprises both the protease as such and a protease produced by a method. All statements regarding the protease therefore relate both to the protease as such and to the proteases produced by means of corresponding methods.

Further aspects relate to the nucleic acids coding for these proteases, to non-human host cells containing proteases or nucleic acids, and to agents comprising proteases, in particular washing and cleaning agents, to washing and cleaning methods, and to uses of the proteases in washing or cleaning agents in order to remove protein-containing stains.

“At least one,” as used herein, means one or more, i.e. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more.

DETAILED DESCRIPTION

Amino acid substitutions at the positions described herein bring about improved storage stability of this altered protease in washing and cleaning agents.

In preferred embodiments of the protease, the protease has substitutions selected from the amino acid substitutions 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q and 224A/G, in particular from 6W, 89A, 131H, 166M, 189T, 211N and 224A, at the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224. In various embodiments, the protease has corresponding substitutions at at least two of these positions. These two substitutions may preferably be those at positions (i) 189 and 224, (ii) 166 and 189 or (iii) 166 and 211 and optionally at least one other position. These are preferably 189T, 224A, 166M and 211N.

In various embodiments, the protease has

- (1) amino acid substitutions, in particular the amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E, at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; and
- (2) amino acid substitutions selected from 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q and 224A/G, preferably selected from 6W, 89A, 131H, 166M, 189T, 211N and 224A, at at least two of the positions corresponding to positions 6, 89, 131, 166, 189, 211 and 224.

In various embodiments, the protease has

- (1) amino acid substitutions, in particular the amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E, at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; and
- (2) at the positions corresponding to positions
 - (a) 166 and 189, the amino acid substitutions 166M and 189T;
 - (b) 166, 189 and 224, the amino acid substitutions 166M, 189T and 224A;
 - (c) 89, 131, 189 and 224, the amino acid substitutions 89A, 131H, 189T and 224A;
 - (d) 89, 189 and 224, the amino acid substitutions 89A, 189T and 224A;
 - (e) 6, 189 and 224, the amino acid substitutions 6W, 189T and 224A;
 - (f) 166 and 211, the amino acid substitutions 166M and 211N;
 - (g) 6 and 189, the amino acid substitutions 6W and 189T;

- (h) 131 and 189, the amino acid substitutions 131H and 189T;
- (i) 89 and 131, the amino acid substitutions 89A and 131H;
- (j) 89 and 166, the amino acid substitutions 89A and 166M;
- (k) 89 and 189, the amino acid substitutions 89A and 189T;
- (l) 89 and 211, the amino acid substitutions 89A and 211N;
- (m) 89 and 224, the amino acid substitutions 89A and 224A;
- (n) 131 and 166, the amino acid substitutions 131H and 166M;
- (o) 131 and 211, the amino acid substitutions 131H and 211N;
- (p) 131 and 224, the amino acid substitutions 131H and 224A; or
- (q) 189 and 224, the amino acid substitutions 189T and 224A.

In various embodiments, the proteases have the amino acid substitutions 9T, 130D, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271 and, at the positions corresponding to positions

- (a) 166 and 189, the amino acid substitutions 166M and 189T;
- (b) 166, 189 and 224, the amino acid substitutions 166M, 189T and 224A;
- (c) 89, 131, 189 and 224, the amino acid substitutions 89A, 131H, 189T and 224A;
- (d) 89, 189 and 224, the amino acid substitutions 89A, 189T and 224A;
- (e) 6, 189 and 224, the amino acid substitutions 6W, 189T and 224A; or
- (f) 166 and 211, the amino acid substitutions 166M and 211N.

In various embodiments, the proteases have the amino acid substitutions 9T, 130V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271 as well as the amino substitutions 89A, 131H, 189T and 224A at the positions corresponding to positions 89, 131, 189 and 224.

In various embodiments, the protease has amino acid substitutions, in particular the amino acid substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E, at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; and at least one, preferably 1, 2, 3, 4, 5 or 6, for example 1, 2, 3 or 4, further amino acid substitution(s) at one or more of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224, said amino acid substitution(s) preferably being selected from: 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q and 224A/G, more preferably from: 6W, 89A, 131H, 166M, 189T, 211N and 224A.

The remainder of the sequence of the above-mentioned proteases has a sufficient sequence identity with SEQ ID NO:1 such that the overall sequence identity of the protease is at least 70%, preferably at least 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96% or 97%. In various embodiments, the remainder of the sequence of the protease, i.e. the sequence with the exception of the positions mentioned herein, which may be mutated is at least 80%, preferably at least 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% identical to the corresponding sequence of SEQ ID NO:1. This means that

the sequence of the protease can correspond to the sequence of SEQ ID NO:1 with the exception of the substitutions mentioned.

In one embodiment, the protease has more than 70% and less than 100% sequence identity with SEQ ID NO:1. Preferably, the protease has more than 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96% or 97% and less than 100% sequence identity with SEQ ID NO:1. The sequence of the protease can correspond to the sequence of SEQ ID NO:1 with the exception of the substitutions mentioned herein.

The proteases have enhanced storage stability. They have increased stability in washing or cleaning agents in comparison with the wild-type enzyme and in particular also with respect to the starting variant of the protease (SEQ ID NO:2), in particular when stored for 3 or more days, 4 or more days, 7 or more days, 10 or more days, 12 or more days, 14 or more days, 21 or more days or 28 or more days. Such performance-enhanced proteases allow improved washing results on proteolytically sensitive stains in various temperature ranges, in particular in a temperature range of from 20° C. to 40° C.

Independently of or in addition to increased storage stability, the proteases may also have increased catalytic activity in washing or cleaning agents. In various embodiments, the proteases may have a proteolytic activity which, based on the wild type and/or an already performance-enhanced reference mutation variant of the protease (SEQ ID NO:1 and/or SEQ ID NO:2), is at least 101%, preferably at least 102%. Such performance-enhanced proteases allow improved washing results on proteolytically sensitive stains in various temperature ranges, in particular in a temperature range of from 20° C. to 40° C.

The proteases exhibit enzymatic activity, i.e. they are capable of hydrolyzing peptides and proteins, in particular in a washing or cleaning agent. A protease is therefore an enzyme which catalyzes the hydrolysis of amide/peptide bonds in protein/peptide substrates and is thus able to cleave proteins or peptides. Furthermore, a protease is preferably a mature protease, i.e. the catalytically active molecule without signal peptide(s) and/or propeptide(s). Unless stated otherwise, the sequences given also each refer to mature (processed) enzymes.

In various embodiments, the protease is a free enzyme. This means that the protease can act directly with all the components of an agent and, if the agent is a liquid agent, that the protease is in direct contact with the solvent of the agent (e.g. water). In other embodiments, an agent may contain proteases that form an interaction complex with other molecules or that contain a "coating." In this case, an individual protease molecule or multiple protease molecules may be separated from the other constituents of the agent by a surrounding structure. Such a separating structure may arise from, but is not limited to, vesicles such as a micelle or a liposome. The surrounding structure may also be a virus particle, a bacterial cell or a eukaryotic cell. In various embodiments, an agent may include cells of *Bacillus pumilus* or *Bacillus subtilis* which express the proteases, or cell culture supernatants of such cells.

In various embodiments, the protease comprises an amino acid sequence which, over its entire length, is at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 90.5%, 91%, 91.5%, 92%, 92.5%, 93%, 93.5%, 94%, 94.5%, 95%, 95.5%, 96%, 96.5% or 97% identical to the amino acid sequence given in SEQ ID NO:1 (where "at

least" refers to each of the given values), and has the amino acid substitutions given above in each case based on the numbering according to SEQ ID NO:1. In an embodiment, the feature whereby a protease has the given substitutions means that it contains one (of the given) substitution(s) at the relevant position, i.e. at least the given positions are not otherwise mutated or deleted, for example by fragmentation of the protease. In various embodiments, the proteases described herein, with the exception of the explicitly mentioned substitutions, have the sequence of SEQ ID NO:1, i.e. apart from the substituted positions, they are 100% identical to the sequence according to SEQ ID NO:1.

The identity of nucleic acid or amino acid sequences is determined by a sequence comparison. This sequence comparison is based on the BLAST algorithm established and commonly used in the prior art (cf. for example Altschul, S. F., Gish, W., Miller, W., Myers, E. W. & Lipman, D. J. (1990) "Basic local alignment search tool." J. Mol. Biol. 215: 403-410, and Altschul, Stephan F., Thomas L. Madden, Alejandro A. Schaffer, Jinghui Zhang, Hheng Zhang, Webb Miller, and David J. Lipman (1997): "Gapped BLAST and PSI-BLAST: a new generation of protein database search programs"; Nucleic Acids Res., 25, pp. 3389-3402) and takes place in principle by similar sequences of nucleotides or amino acids in the nucleic acid or amino acid sequences being assigned to one another. A tabular association of the positions concerned is referred to as alignment. Another algorithm available in the prior art is the FASTA algorithm. Sequence comparisons (alignments), in particular multiple sequence comparisons, are created using computer programs. The Clustal series (cf. for example Chenna et al. (2003): Multiple sequence alignment with the Clustal series of programs. Nucleic Acid Research 31, 3497-3500), T-Coffee (cf. for example Notredame et al. (2000): T-Coffee: A novel method for multiple sequence alignments. J. Mol. Biol. 302, 205-217) or programs based on these programs or algorithms are often used, for example. Sequence comparisons (alignments) using the computer program Vector NTI® Suite 10.3 (Invitrogen Corporation, 1600 Faraday Avenue, Carlsbad, California, USA) with the predetermined default parameters, and the AlignX module of which for sequence comparisons is based on ClustalW, are also possible. Unless stated otherwise, the sequence identity given herein is determined by the BLAST algorithm.

Such a comparison also allows conclusions to be drawn regarding the similarity of the compared sequences. It is usually given in percent identity, i.e. the proportion of identical nucleotides or amino acid residues in said sequences or in an alignment of corresponding positions. The broader concept of homology takes conserved amino acid exchanges into account in the case of amino acid sequences, i.e. amino acids having similar chemical activity, since they usually perform similar chemical activities within the protein. Therefore, the similarity between the compared sequences can also be expressed in percent homology or percent similarity. Identity and/or homology information can be provided regarding whole polypeptides or genes or only regarding individual regions. Homologous or identical regions of different nucleic acid or amino acid sequences are therefore defined by matches in the sequences. Such regions often have identical functions. They can be small and comprise only a few nucleotides or amino acids. Often, such small regions perform essential functions for the overall activity of the protein. It may therefore be expedient to relate sequence matches only to individual, optionally small, regions. Unless stated otherwise, however, identity or

homology information in the present application relates to the entire length of the particular nucleic acid or amino acid sequence indicated.

In an embodiment, the indication that an amino acid position corresponds to a numerically designated position in SEQ ID NO:1 therefore means that the corresponding position is associated with the numerically designated position in SEQ ID NO:1 in an alignment as defined above.

In a further embodiment, the protease is characterized in that the cleaning performance thereof (after storage, e.g. over 4 weeks) is not significantly reduced compared with that of a protease comprising an amino acid sequence that corresponds to the amino acid sequence given in SEQ ID NO:2, i.e. has at least 80% of the reference washing performance, preferably at least 100%, more preferably at least 110% or more. The cleaning performance can be determined in a washing system containing a washing agent in a dosage between 4.5 and 7.0 grams per liter of washing liquor, and the protease, the proteases to be compared being used in the same concentration (based on active protein), and the cleaning performance with respect to a stain on cotton being determined by measuring the degree of cleaning of the washed textiles. For example, the washing process can take place for 60 minutes at a temperature of 40° C. and the water can have a water hardness between 15.5° dH and 16.5° dH (German hardness). The concentration of the protease in the washing agent intended for this washing system is 0.001 to 0.1 wt. %, preferably 0.01 to 0.06 wt. % based on active, purified protein.

A liquid reference washing agent for such a washing system may be composed as follows (all figures in wt. %): 4.4% alkyl benzene sulfonic acid, 5.6% further anionic surfactants, 2.4% C₁₂-C₁₈ Na salts of fatty acids (soaps), 4.4% non-ionic surfactants, 0.2% phosphonates, 1.4% citric acid, 0.95% NaOH, 0.01% defoamer, 2% glycerol, 0.08% preservatives, 1% ethanol, and the remainder being demineralized water. Preferably, the dosage of the liquid washing agent is between 4.5 and 6.0 grams per liter of washing liquor, for example 4.7, 4.9 or 5.9 grams per liter of washing liquor. Washing in a pH range between pH 7 and pH 10.5, preferably between pH 7.5 and pH 8.5, is preferred.

In an embodiment, the cleaning performance is determined for example at 20° C. or 40° C. using a liquid washing agent as stated above, the washing process preferably being carried out for 60 minutes at 600 rpm.

The degree of whiteness, i.e. the lightening of stains, as a measure of the cleaning performance is determined by optical measuring methods, preferably photometrically. A suitable device for this purpose is for example the Minolta CM508d spectrometer. Usually, the devices used for the measurement are calibrated beforehand with a white standard, preferably a supplied white standard.

The activity-equivalent use of the relevant protease ensures that the respective enzymatic properties, for example the cleaning performance on certain stains, are compared even if the ratio of active substance to total protein (the values of the specific activity) diverges. In general, a low specific activity can be compensated for by adding a larger amount of protein.

Otherwise, methods for determining protease activity are well known to, and routinely used by, a person skilled in the art of enzyme technology. For example, such methods are disclosed in Tenside, vol. 7 (1970), pp. 125-132. Alternatively, the protease activity can be determined by the release of the chromophore para-nitroaniline (pNA) from the substrate suc-L-Ala-L-Ala-L-Pro-L-Phe-p-nitroanilide (AAPF). The protease cleaves the substrate and releases

pNA. The release of the pNA causes an increase in absorbance at 410 nm, the temporal progression of which is a measure of the enzymatic activity (cf. Del Mar et al., 1979). The measurement is carried out at a temperature of 25° C., a pH of 8.6, and a wavelength of 410 nm. The measuring time is 5 min and the measuring interval is 20 s to 60 s. The protease activity is usually indicated in protease units (PE). Suitable protease activities amount to 2.25, 5 or 10 PE per mL of washing liquor, for example. However, the protease activity is not equal to zero.

An alternative test for establishing the proteolytic activity of the proteases is an optical measuring method, preferably a photometric method. The appropriate test involves the protease-dependent cleavage of the substrate protein casein. This is cleaved by the protease into a multitude of smaller partial products. The totality of these partial products has an increased absorption at 290 nm compared with uncleaved casein, it being possible for this increased absorption to be determined using a photometer, and thus for a conclusion to be drawn regarding the enzymatic activity of the protease.

The protein concentration can be determined using known methods, for example the BCA method (bicinchoninic acid; 2,2'-bichinoly-4,4'-dicarboxylic acid) or the Biuret method (A. G. Gornall, C. S. Bardawill and M. M. David, J. Biol. Chem., 177 (1948), p. 751-766). The active protein concentration can be determined in this regard by titrating the active centers using a suitable irreversible inhibitor and determining the residual activity (cf. M. Bender et al., J. Am. Chem. Soc. 88, 24 (1966), pp. 5890-5913).

In addition to the amino acid alterations discussed above, proteases can have other amino acid alterations, in particular amino acid substitutions, insertions or deletions. Such proteases are, for example, developed by targeted genetic alteration, i.e. by mutagenesis methods, and optimized for specific applications or with regard to specific properties (for example with regard to their catalytic activity, stability, etc.). Furthermore, nucleic acids can be introduced into recombination approaches and can thus be used to generate completely new types of proteases or other polypeptides.

The aim is to introduce targeted mutations such as substitutions, insertions or deletions into the known molecules in order, for example, to improve the cleaning performance of enzymes. For this purpose, in particular the surface charges and/or the isoelectric point of the molecules and thus their interactions with the substrate can be altered. For instance, the net charge of the enzymes can be altered in order to influence the substrate binding, in particular for use in washing and cleaning agents. Alternatively or additionally, one or more corresponding mutations can increase the stability or catalytic activity of the protease and thus improve its cleaning performance. Advantageous properties of individual mutations, e.g. individual substitutions, can complement one another. A protease which has already been optimized with regard to specific properties, for example with respect to its stability during storage, can therefore also be developed.

For the description of substitutions relating to exactly one amino acid position (amino acid exchanges), the following convention is used herein: first, the naturally occurring amino acid is designated in the form of the internationally used one-letter code, followed by the associated sequence position and finally the inserted amino acid. Several or alternative exchanges within the same polypeptide chain are separated by slashes. "130D/V" thus means that position 130 has mutated to D or V. For insertions, additional amino acids are named following the sequence position. In the case of deletions, the missing amino acid is replaced by a symbol,

for example a star or a dash, or a Δ is indicated before the corresponding position. For example, P9T describes the substitution of proline at position 9 by threonine, P9TH describes the insertion of histidine following the amino acid threonine at position 9 and P9* or Δ P9 describes the deletion of proline at position 9. This nomenclature is known to a person skilled in the field of enzyme technology.

A protease may be obtainable from a protease as described above as the starting molecule by single or multiple conservative amino acid substitution, the protease in the numbering according to SEQ ID NO:1 having the above-described amino acid substitutions. The term "conservative amino acid substitution" means the exchange (substitution) of one amino acid residue for another amino acid residue, with this exchange not resulting in a change to the polarity or charge at the position of the exchanged amino acid, e.g. the exchange of a nonpolar amino acid residue for another nonpolar amino acid residue. Conservative amino acid substitutions include, for example: G=A=S, I=V=L=M, D=E, N=Q, K=R, Y=F, S=T, G=A=I=V=L=M=Y=F=W=P=S=T.

Alternatively or in addition, the protease is characterized in that it is obtainable from a protease as a starting molecule by fragmentation or deletion, insertion or substitution mutagenesis, and comprises an amino acid sequence which matches the starting molecule over a length of at least 200, 210, 220, 230, 240, 250, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, or 274 contiguous amino acids, the above-described amino acid substitutions, i.e. the substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271, and a further amino acid substitution at at least one, preferably at least two, of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224, still being present.

For instance, it is possible to delete individual amino acids at the termini or in the loops of the enzyme without the proteolytic activity being lost or diminished in the process. Furthermore, such fragmentation or deletion, insertion or substitution mutagenesis can also for example reduce the allergenicity of the enzymes concerned and thus improve their overall applicability. Advantageously, the enzymes retain their proteolytic activity even after mutagenesis, i.e. their proteolytic activity corresponds at least to that of the starting enzyme, i.e. in a preferred embodiment the proteolytic activity is at least 80%, preferably at least 90% of the activity of the starting enzyme. Other substitutions can also exhibit advantageous effects. Both single and multiple contiguous amino acids can be exchanged for other amino acids.

The further amino acid positions are in this case defined by an alignment of the amino acid sequence of a protease with the amino acid sequence of the protease from *Bacillus pumilus*, as given in SEQ ID NO:1. Furthermore, the assignment of the positions depends on the mature protein. This assignment is also to be used in particular if the amino acid sequence of a protease comprises a higher number of amino acid residues than the protease from *Bacillus pumilus* according to SEQ ID NO:1. Proceeding from the above-mentioned positions in the amino acid sequence of the protease from *Bacillus pumilus*, the alteration positions in a protease are those which are assigned to precisely these positions in an alignment.

Advantageous positions for sequence alterations, in particular substitutions, of the protease from *Bacillus pumilus*, which are of particular significance when transferred to homologous positions of the proteases and which impart advantageous functional properties to the protease, are

therefore the positions which correspond to the positions described herein in an alignment, i.e. in the numbering according to SEQ ID NO: 1. At the positions mentioned, the following amino acid residues are present in the wild-type molecule of the protease from *Bacillus pumilus*: P9, N130, T133, N144, Y217, N252 and Q271 as well as Y6, S89, G131, G166, S189, S211 and S224.

Further confirmation of the correct assignment of the amino acids to be altered, i.e. in particular their functional correspondence, can be provided by comparative experiments, according to which the two positions assigned to one another on the basis of an alignment are altered in the same way in both compared proteases, and observations are made as to whether the enzymatic activity is altered in the same way in both cases. If, for example, an amino acid exchange in a specific position of the protease from *Bacillus pumilus* according to SEQ ID NO:1 is accompanied by an alteration of an enzymatic parameter, for example an increase in the KM value, and a corresponding alteration of the enzymatic parameter, for example likewise an increase in the KM value, is observed in a protease variant of which the amino acid exchange has been achieved by the same introduced amino acid, this can therefore be considered to be confirmation of the correct assignment.

All of these aspects are also applicable to the methods for producing a protease. Accordingly, a method further comprises one or more of the following method steps:

- introducing single or multiple conservative amino acid substitution into the protease, the protease comprising the substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; and at least one further amino acid substitution at at least one or at least two of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224;
- altering the amino acid sequence by fragmentation or deletion, insertion or substitution mutagenesis such that the protease comprises an amino acid sequence which matches the starting molecule over a length of at least 200, 210, 220, 230, 240, 250, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273 or 274 contiguous amino acids, the protease comprising the substitutions 9T, 130D/V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; and at least one further amino acid substitution at at least one or at least two of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224.

All embodiments also apply to the methods.

In further embodiments, the protease or the protease produced by means of a method is still at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 90.5%, 91%, 91.5%, 92%, 92.5%, 93%, 93.5%, 94%, 94.5%, 95%, 95.5%, 96%, 96.5% or 97% identical to the amino acid sequence given in SEQ ID NO:1 over its entire length. Alternatively, the protease or the protease produced by means of a method is still at least 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 90.5%, 91%, 91.5%, 92%, 92.5%, 93%, 93.5%, 94%, 94.5%, 95%, 95.5%, 96%, 96.5%, 97%, 97.5%, 98%, 98.5% or 99% identical to the amino acid sequences given in SEQ ID NO:2 over its entire length. The protease or the protease produced by means of a method has the amino acid substitutions from 9T, 130D/V, 133A, 144K, 217M, 252T and 271E at the positions corresponding to positions 9, 130, 133, 144, 252

and 271; and at least one, preferably at least two, further amino acid substitution(s) at at least one of the positions corresponding to positions 6, 89, 131, 166, 189, 211 or 224, in each case based on the numbering according to SEQ ID NO:1. Examples thereof are the following amino acid substitution variants: P9T, N130D, T133A, N144K, Y217M, N252T and Q271E combined with one of (i) G166M and S189T; (ii) G166M, S189T and S224A (iii) S89A, G131H and S189T; (iv) S89A and S189T; (v) Y6W, S189T and S224A; or (vi) G166M and S211N, or P9T, N130V, T133A, N144K, Y217M, N252T and Q271E combined with S89A, G131H and S189T, the numbering being based in each case on the numbering according to SEQ ID NO:1 and the variants described in the examples.

A protease described above which may be stabilized, in particular by one or more mutations, for example substitutions, or by coupling to a polymer. An increase in stability during storage and/or during use, for example in the washing process, leads to longer enzymatic activity and thus improves the cleaning performance. In principle, all stabilization options which are described in the prior art and/or are appropriate are considered. Those stabilizations are preferred which are achieved by mutations of the enzyme itself, since such stabilizations do not require any further work steps following the recovery of the enzyme. Examples of sequence alterations suitable for this purpose are mentioned above. Further suitable sequence alterations are known from the prior art.

Further possibilities for stabilization are, for example:

altering the binding of metal ions, in particular the calcium binding sites, for example by exchanging one or more of the amino acid(s) that are involved in the calcium binding with one or more negatively charged amino acids and/or by introducing sequence alterations in at least one of the sequences of the two amino acids arginine and glycine;

protecting against the influence of denaturing agents such as surfactants by mutations that cause an alteration of the amino acid sequence on or at the surface of the protein;

exchanging amino acids near the N-terminus with those likely to contact the rest of the molecule via non-covalent interactions, thus contributing to the maintenance of the globular structure.

Preferred embodiments are those in which the enzyme is stabilized in several ways, as several stabilizing mutations act additively or synergistically.

A protease as described above may have at least one chemical modification. A protease with such an alteration is referred to as a derivative, i.e. the protease is derivatized.

In the context of the present application, derivatives are thus understood to mean those proteins of which the pure amino acid chain has been chemically modified. Such derivatizations can be achieved, for example, in vivo by the host cell that expresses the protein. In this regard, couplings of low-molecular-weight compounds such as lipids or oligosaccharides are particularly noteworthy. However, the derivatizations may also be carried out in vitro, for example by the chemical conversion of a side chain of an amino acid or by covalent bonding of another compound to the protein. For example, it is possible to couple amines to carboxyl groups of an enzyme in order to alter the isoelectric point. Another such compound may also be another protein that is bound to a protein via bifunctional chemical compounds, for example. Derivatization is also understood to mean the covalent bonding to a macromolecular carrier or a non-covalent inclusion in suitable macromolecular cage struc-

tures. Derivatizations may, for example, affect the substrate specificity or bonding strength to the substrate or cause a temporary blockage of the enzymatic activity when the coupled substance is an inhibitor. This can be expedient, for example, for the period of storage. Such modifications may further affect the stability or enzymatic activity. They can also be used to reduce the allergenicity and/or immunogenicity of the protein and thus, for example, increase its skin compatibility. For example, couplings with macromolecular compounds, for example polyethylene glycol, can improve the protein in terms of stability and/or skin compatibility.

Derivatives of a protein can also be understood in the broadest sense to mean preparations of these proteins. Depending on the recovery, processing or preparation, a protein can be combined with various other substances, for example from the culture of the producing microorganisms. A protein may also have been deliberately added to other substances, for example to increase its storage stability. This is also irrespective of whether or not it actually exhibits this enzymatic activity in a particular preparation. This is because it may be desired that it has no or only low activity during storage, and exhibits its enzymatic function only at the time of use. This can be controlled via appropriate accompanying substances, for example. In particular, the joint preparation of proteases with specific inhibitors is possible in this regard.

Of all the proteases or protease variants and/or derivatives described above, those of which the storage stability corresponds at least to the protease according to SEQ ID NO:2 or the variants tested in the examples, and/or of which the cleaning performance corresponds at least to the protease according to SEQ ID NO:2 or the variants tested in the examples, are particularly preferred, the cleaning performance being determined in a washing system as described above.

A nucleic acid may code for a protease, as well as to a vector containing such a nucleic acid, in particular a cloning vector or an expression vector.

These may be DNA or RNA molecules. They can be present as a single strand, as a single strand that is complementary to this single strand, or as a double strand. In particular in the case of DNA molecules, the sequences of the two complementary strands must be taken into account in all three possible reading frames. Furthermore, it should be noted that different codons, i.e. base triplets, can code for the same amino acids such that a particular amino acid sequence can be coded by a plurality of different nucleic acids. Due to this degeneracy of the genetic code, all of the nucleic acid sequences which can code any of the proteases described above are included herein. A person skilled in the art is able to determine these nucleic acid sequences unequivocally since, despite the degeneracy of the genetic code, defined amino acids can be assigned to individual codons. Therefore, a person skilled in the art proceeding from an amino acid sequence can easily determine nucleic acids coding for said amino acid sequence. Furthermore, in the case of nucleic acids, one or more codons may be replaced by synonymous codons. This aspect relates in particular to the heterologous expression of the enzymes. For instance, every organism, for example a host cell of a production strain, has a particular codon usage. Codon usage is understood to mean the translation of the genetic code into amino acids by the relevant organism. Bottlenecks can occur in the protein biosynthesis if the codons on the nucleic acid in the organism are faced with a comparatively small number of loaded tRNA molecules. Although coding for the same amino acid, this results in a codon being translated less

efficiently in the organism than a synonymous codon coding for the same amino acid. Due to the presence of a higher number of tRNA molecules for the synonymous codon, it can be translated more efficiently in the organism.

Using methods which are currently generally known, such as chemical synthesis or the polymerase chain reaction (PCR), in conjunction with molecular-biological and/or protein-chemical standard methods, it is possible for a person skilled in the art to produce the corresponding nucleic acids and even complete genes on the basis of known DNA and/or amino acid sequences. Such methods are known for example from Sambrook, J., Fritsch, E. F. and Maniatis, T. 2001. Molecular cloning: a laboratory manual, 3rd edition, Cold Spring Laboratory Press.

As described herein, vectors are understood to mean elements consisting of nucleic acids, which elements contain a nucleic acid as the characteristic nucleic acid region. They are able to establish these as a stable genetic element in a species or cell line over several generations or cell divisions. Vectors are special plasmids, i.e. circular genetic elements, in particular when used in bacteria. As described herein, a nucleic acid is cloned into a vector. The vectors include, for example, those originating from bacterial plasmids, viruses or bacteriophages, or predominantly synthetic vectors or plasmids with elements of a wide variety of origins. With the additional genetic elements present in each case, vectors are able to establish themselves as stable units in the corresponding host cells over several generations. They may be present as separate units in an extrachromosomal manner or integrated into a chromosome or chromosomal DNA.

Expression vectors comprise nucleic acid sequences which enable them to replicate in the host cells containing them, preferably microorganisms, particularly preferably bacteria, and to express a contained nucleic acid there. The expression is in particular influenced by the promoter(s) that regulate the transcription. In principle, the expression can take place by the natural promoter originally located before the nucleic acid to be expressed, but also by a promoter of the host cell provided on the expression vector or also by a modified or completely different promoter of another organism or another host cell. In the present case, at least one promoter is provided for the expression of a nucleic acid and used for the expression thereof. Furthermore, expression vectors can be regulatable, for example by changing the cultivation conditions or when a specific cell density of the host cells containing them is reached or by addition of specific substances, in particular activators of gene expression. An example of such a substance is the galactose derivative isopropyl-O-D-thiogalactopyranoside (IPTG), which is used as an activator of the bacterial lactose operon (lac operon). In contrast with expression vectors, the nucleic acid contained is not expressed in cloning vectors.

A non-human host cell may include a nucleic acid or a vector or which contains a protease, in particular one which secretes the protease into the medium surrounding the host cell. Preferably, a nucleic acid or a vector is transformed into a microorganism, which then represents a host cell. Alternatively, individual components, i.e. nucleic acid parts or fragments of a nucleic acid, can be introduced into a host cell such that the resulting host cell contains a nucleic acid or a vector. This procedure is particularly suitable when the host cell already contains one or more constituents of a nucleic acid or a vector and the further constituents are then supplemented accordingly. Methods for transforming cells are established in the prior art and are well known to a person skilled in the art. In principle, all cells, i.e. prokaryotic or eukaryotic cells, are suitable as host cells. Host cells that can

be managed in a genetically advantageous manner, for example in terms of the transformation with the nucleic acid or the vector and the stable establishment thereof, are preferred, for example unicellular fungi or bacteria. Furthermore, preferred host cells are characterized by good microbiological and biotechnological manageability. This relates, for example, to easy cultivation, high growth rates, low requirements for fermentation media and good production and secretion rates for foreign proteins. Preferred host cells secrete the (transgenically) expressed protein into the medium surrounding the host cells. Furthermore, the proteases can be modified by the cells producing them after their production, for example by attachment of sugar molecules, formylations, aminations, etc. Such post-translational modifications can functionally influence the protease.

Other preferred embodiments are those host cells of which the activity can be regulated on account of genetic regulatory elements, which are, for example, made available on the vector but may also be present in these cells from the outset. These host cells may be induced to express for example by the controlled addition of chemical compounds which are used as activators, by changing the cultivation conditions, or when a specific cell density is reached. This enables economical production of the proteins. An example of such a compound is IPTG, as described above.

Prokaryotic or bacterial cells are preferred host cells. Bacteria are characterized by short generation times and low demands on cultivation conditions. As a result, cost-effective cultivation methods or production methods can be established. In addition, a person skilled in the art has a wealth of experience in the case of bacteria in fermentation technology. For a specific production, gram-negative or gram-positive bacteria may be suitable for a wide variety of reasons to be determined experimentally in individual cases, such as nutrient sources, product formation rate, time requirement, etc.

In the case of gram-negative bacteria, such as *Escherichia coli*, a large number of proteins are secreted into the periplasmic space, i.e. into the compartment between the two membranes enclosing the cells. This may be advantageous for particular applications. Furthermore, gram-negative bacteria can also be designed such that they eject the expressed proteins not only into the periplasmic space, but into the medium surrounding the bacterium. In contrast, gram-positive bacteria such as bacilli or actinomycetes or other representatives of *Actinomycetales* have no outer membrane, and therefore secreted proteins are released immediately into the medium surrounding the bacteria, usually the nutrient medium, from which the expressed proteins can be purified. They can be isolated directly from the medium or further processed. In addition, gram-positive bacteria are related or identical to most of the origin organisms for technically significant enzymes and usually even form comparable enzymes, meaning that they have a similar codon usage and the protein synthesizer is naturally aligned accordingly.

Host cells may be altered in terms of their requirements for the culture conditions, may have different or additional selection markers or may express other or additional proteins. In particular, this may also involve those host cells which transgenically express several proteins or enzymes.

The present disclosure is applicable in principle to all microorganisms, in particular to all fermentable microorganisms, particularly preferably those of the genus *Bacillus*, and leads to it being possible to produce proteins by the use of such microorganisms. Such microorganisms then represent host cells.

In a further embodiment, the host cell is characterized in that it is a bacterium, preferably one selected from the group of the genera of *Escherichia*, *Klebsiella*, *Bacillus*, *Staphylococcus*, *Corynebacterium*, *Arthrobacter*, *Streptomyces*, *Stenotrophomonas* and *Pseudomonas*, more preferably one selected from the group of *Escherichia coli*, *Klebsiella planticola*, *Bacillus licheniformis*, *Bacillus lentus*, *Bacillus amyloliquefaciens*, *Bacillus subtilis*, *Bacillus alcalophilus*, *Bacillus globigii*, *Bacillus gibsonii*, *Bacillus clausii*, *Bacillus halodurans*, *Bacillus pumilus*, *Staphylococcus carnosus*, *Corynebacterium glutamicum*, *Arthrobacter oxidans*, *Streptomyces lividans*, *Streptomyces coelicolor* and *Stenotrophomonas maltophilia*.

The host cell may also be a eukaryotic cell, however, which is characterized in that it has a cell nucleus. A host cell has a cell nucleus. In contrast with prokaryotic cells, eukaryotic cells are capable of post-translationally modifying the protein formed. Examples thereof are fungi such as actinomycetes or yeasts such as *Saccharomyces* or *Kluyveromyces*. This can be particularly advantageous, for example, if the proteins are to undergo specific modifications in connection with their synthesis that make such systems possible. Modifications carried out by eukaryotic systems, in particular in connection with the protein synthesis, include, for example, the binding of low-molecular-weight compounds such as membrane anchors or oligosaccharides. Such oligosaccharide modifications may be desirable, for example, to lower the allergenicity of an expressed protein. Co-expression with the enzymes naturally formed by such cells, such as cellulases, may be advantageous. Furthermore, for example, thermophilic fungal expression systems may be particularly suitable for the expression of temperature-resistant proteins or variants.

The host cells are cultivated and fermented in the conventional way, for example in discontinuous or continuous systems. In the first case, a suitable nutrient medium is inoculated with the host cells and the product is harvested from the medium after a period to be determined experimentally. Continuous fermentations are characterized by the achievement of a flow equilibrium, in which cells partially die over a comparatively long period of time but also grow back and the protein formed can be removed from the medium at the same time.

- a) Host cells are preferably used to produce proteases. A method for producing a protease may include cultivating a host cell, and
- b) isolating the protease from the culture medium or from the host cell.

This subject matter preferably comprises fermentation processes. Fermentation processes are known per se from the prior art and represent the actual large-scale production step, usually followed by a suitable purification method of the prepared product, for example the proteases. All fermentation processes which are based on a corresponding method for producing a protease may be used in the methods herein.

Fermentation processes which are characterized in that the fermentation is carried out via a feed strategy shall be considered in particular. In this case, the media constituents that are consumed by the continuous cultivation are added. As a result, considerable increases can be achieved both in the cell density and in the cell mass or dry mass and/or in particular in the activity of the protease of interest. Furthermore, the fermentation can also be designed in such a way that undesired metabolic products are filtered out or neutralized by adding buffers or suitable counter ions.

The produced protease can be harvested from the fermentation medium. Such a fermentation process is preferable to

isolation of the protease from the host cell, i.e. product preparation from the cell mass (dry matter), but requires the provision of suitable host cells or one or more suitable secretion markers or mechanisms and/or transport systems for the host cells to secrete the protease into the fermentation medium. Without secretion, the protease can alternatively be isolated from the host cell, i.e. purified from the cell mass, for example by precipitation with ammonium sulfate or ethanol, or by chromatographic purification.

All of the above-mentioned aspects can be combined into methods in order to produce a protease.

An agent may include a protease as described above. The agent is preferably a washing or cleaning agent.

All conceivable types of washing or cleaning agents may be used, both concentrates and undiluted agents, for use on a commercial scale, in washing machines or for hand washing or cleaning. These include, for example, washing agents for textiles, carpets or natural fibers, for which the term washing agent is used. These include, for example, dishwashing detergents for dishwashers or manual dishwashing detergents or cleaners for hard surfaces such as metal, glass, porcelain, ceramics, tiles, stone, painted surfaces, plastics, wood or leather, for which the term cleaning agent is used, i.e. in addition to manual and mechanical dishwashing detergents, also, for example, scouring agents, glass cleaners, WC toilet scenters, etc. The washing and cleaning agents also include auxiliary washing agents which are added to the actual washing agent during manual or mechanical textile washing in order to achieve a further effect. Furthermore, washing and cleaning agents also include textile pre-treatment and post-treatment agents, i.e. those agents with which the item of laundry is brought into contact before the actual washing cycle, for example to loosen stubborn soiling, and also those agents which give the laundry further desirable properties such as a pleasant feel, crease resistance or low static charge in a step subsequent to the actual textile wash. Inter alia, softeners are included in the last-mentioned agents.

The washing or cleaning agents, which may be in the form of powdered solids, in further-compacted particulate form, as homogeneous solutions or suspensions, may contain, in addition to a protease, all known ingredients conventional in such agents, with preferably at least one other ingredient being present in the agent. The agents may in particular contain surfactants, builders, peroxygen compounds or bleach activators. They may also contain water-miscible organic solvents, further enzymes, sequestering agents, electrolytes, pH regulators and/or further auxiliaries such as optical brighteners, graying inhibitors, foam regulators, as well as dyes and fragrances, and combinations thereof.

In particular, a combination of a protease with one or more further ingredients of the agent is advantageous, since, in preferred embodiments, such an agent has improved cleaning performance by virtue of resulting synergisms. In particular, combining a protease with a surfactant and/or a builder and/or a peroxygen compound and/or a bleach activator can result in such a synergism. However, in preferred embodiments, the agent may not contain boric acid.

Advantageous ingredients of agents are disclosed in international patent application WO 2009/121725, starting at the penultimate paragraph of page 5 and ending after the second paragraph on page 13. Reference is expressly made to this disclosure and the disclosure therein is incorporated in the present patent application by reference.

An agent advantageously contains the protease in an amount of from 2 µg to 20 mg, preferably from 5 µg to 17.5 mg, more preferably from 20 µg to 15 mg and most

particularly preferably from 50 μg to 10 mg per g of the agent. In various embodiments, the concentration of the protease (active enzyme) described herein in the agent is >0 to 1 wt. %, preferably 0.0001 or 0.001 to 0.1 wt. %, based on the total weight of the agent or composition. Further, the protease contained in the agent, and/or other ingredients of the agent, may be coated with a substance which is impermeable to the enzyme at room temperature or in the absence of water, and which becomes permeable to the enzyme under conditions of use of the agent. Such an embodiment is thus characterized in that the protease is coated with a substance which is impermeable to the protease at room temperature or in the absence of water. Furthermore, the washing or cleaning agent itself may also be packaged in a container, preferably an air-permeable container, from which it is released shortly before use or during the washing process.

In further embodiments, the agent is characterized in that it

- (a) is present in solid form, in particular as a flowable powder having a bulk density of from 300 g/L to 1200 g/L, in particular from 500 g/L to 900 g/L, or
- (b) is present in pasty or liquid form, and/or
- (c) is present in the form of a gel or in the form of dosing pouches, and/or
- (d) is present as a single-component system, or
- (e) is divided into a plurality of components.

These embodiments include all solid, powdered, liquid, gel or pasty administration forms of agents, which may optionally also consist of a plurality of phases and can be present in compressed or uncompressed form. The agent may be present as a flowable powder, in particular having a bulk density of from 300 g/L to 1200 g/L, in particular from 500 g/L to 900 g/L or from 600 g/L to 850 g/L. The solid administration forms of the agent also include extrudates, granules, tablets or pouches. Alternatively, the agent may also be in liquid, gel or pasty form, for example in the form of a non-aqueous liquid washing agent or a non-aqueous paste or in the form of an aqueous liquid washing agent or a water-containing paste. Liquid agents are generally preferred. The agent may also be present as a single-component system. Such agents consist of one phase. Alternatively, an agent may also consist of a plurality of phases. Such an agent is therefore divided into a plurality of components.

Washing or cleaning agents may contain only one protease. Alternatively, they may also contain other hydrolytic enzymes or other enzymes in a concentration that is expedient for the effectiveness of the agent. A further embodiment is therefore represented by agents which further comprise one or more further enzymes. Further enzymes which can preferably be used are all enzymes which can exhibit catalytic activity in the agent, in particular a lipase, amylase, cellulase, hemicellulase, mannanase, tannase, xylanase, xanthanase, xyloglucanase, β -glucosidase, pectinase, carrageenase, perhydrolase, oxidase, oxidoreductase or another protease, which is different from the proteases, as well as mixtures thereof. Further enzymes are contained in the agent advantageously in an amount of from 1×10^{-8} to 5 wt. % in each case, based on the active protein. Increasingly preferably, each further enzyme is contained in agents in an amount of from 1×10^{-7} to 3 wt. %, from 0.00001 to 1 wt. %, from 0.00005 to 0.5 wt. %, from 0.0001 to 0.1 wt. % and particularly preferably from 0.0001 to 0.05 wt. %, based on active protein. Particularly preferably, the enzymes exhibit synergistic cleaning performance against specific stains or spots, i.e. the enzymes contained in the agent composition support one another in their cleaning performance. Very particularly preferably, there is such synergism between the

protease contained and a further enzyme of an agent, including in particular between said protease and an amylase and/or a lipase and/or a mannanase and/or a cellulase and/or a pectinase. Synergistic effects can arise not only between different enzymes, but also between one or more enzymes and other ingredients of the agent.

In the cleaning agents described herein, the enzymes to be used may furthermore be formulated together with accompanying substances, for example from fermentation. In liquid formulations, the enzymes are preferably used as enzyme liquid formulations.

The enzymes are generally not provided in the form of pure protein, but rather in the form of stabilized, storable and transportable preparations. These ready-made preparations include, for example, the solid preparations obtained through granulation, extrusion, or lyophilization or, particularly in the case of liquid or gel agents, solutions of the enzymes, which are advantageously maximally concentrated, have a low water content, and/or are supplemented with stabilizers or other auxiliaries.

Alternatively, the enzymes can also be encapsulated, for both the solid and the liquid administration form, for example by spray-drying or extrusion of the enzyme solution together with a preferably natural polymer or in the form of capsules, for example those in which the enzymes are enclosed in a set gel, or in those of the core-shell type, in which an enzyme-containing core is coated with a water-, air-, and/or chemical-impermeable protective layer. Further active ingredients such as stabilizers, emulsifiers, pigments, bleaching agents, or dyes can additionally be applied in overlaid layers. Such capsules are applied using inherently known methods, for example by shaking or roll granulation or in fluidized bed processes. Such granules are advantageously low in dust, for example due to the application of polymeric film-formers, and stable in storage due to the coating.

Moreover, it is possible to formulate two or more enzymes together, such that a single granule exhibits a plurality of enzyme activities.

The enzymes can also be incorporated in water-soluble films, such as those used in the formulation of washing and cleaning agents in a unit dosage form. Such a film allows the release of the enzymes following contact with water. As used herein, "water-soluble" refers to a film structure that is preferably completely water-soluble. Preferably, such a film consists of (fully or partially hydrolyzed) polyvinyl alcohol (PVA).

A method for cleaning textiles or hard surfaces may be used in at least one method step, or in that a protease becomes catalytically active in at least one method step, in particular such that the protease is used in an amount of from 40 μg to 4 g, preferably from 50 μg to 3 g, particularly preferably from 100 μg to 2 g, and most particularly preferably from 200 μg to 1 g or in the concentrations described herein.

In various embodiments, the method described above is characterized in that the protease is used at a temperature of from 0 to 100° C., preferably 0 to 60° C., more preferably 20 to 40° C. and most preferably at 25° C.

These include both manual and mechanical methods, with mechanical methods being preferred. Methods for cleaning textiles are generally characterized by the fact that, in a plurality of method steps, various cleaning-active substances are applied to the material to be cleaned and washed off after the exposure time, or in that the material to be cleaned is otherwise treated with a washing agent or a solution or dilution of this agent. The same applies to methods for

19

cleaning all materials other than textiles, in particular hard surfaces. All conceivable washing or cleaning methods can be enhanced in at least one of the method steps by the use of a washing or cleaning agent or a protease, and therefore represent embodiments. All aspects, objects and embodiments described for the protease and agents containing it are also applicable to this subject matter. Therefore, reference is expressly made at this point to the disclosure at the appropriate point with the note that this disclosure also applies to the above-described methods.

Since proteases naturally already have hydrolytic activity and also exhibit this in media which otherwise have no cleaning power, for example in a simple buffer, a single and/or the sole step of such a method can consist in a protease, which is the only cleaning-active component, being brought into contact with the stain, preferably in a buffer solution or in water. This constitutes a further embodiment.

Alternative embodiments are also represented by methods for treating textile raw materials or for textile care, in which a protease becomes active in at least one method step. Among these, methods for textile raw materials, fibers or textiles with natural constituents are preferred, and very particularly for those with wool or silk.

Finally, the proteases described herein in washing or cleaning agents, for example as described above, may be used for the (improved) removal of protein-containing stains, for example from textiles or hard surfaces. In preferred embodiments of this use, the protease in the washing or cleaning agent is stored for 3 or more days, 4 or more days, 7 or more days, 10 or more days, 12 or more days, 14 or more days, 21 or more days or 28 or more days before a washing or cleaning process.

All aspects, objects and embodiments described for the protease and agents containing it are also applicable herein. Therefore, reference is expressly made at this point to the disclosure at the appropriate point with the note that this disclosure also applies to the above-described use.

EXAMPLES

Overview of the Mutations:

A subtilisin-type alkaline protease was derived from *Bacillus pumilus*. From this protease (wild-type *Bacillus pumilus* DSM18097 protease according to SEQ ID NO:1), variants were produced by random mutagenesis, which were then screened, inter alia for improved washing performance and/or enzyme stability. In this way, a performance- and stability-enhanced mutant (C0C [SEQ ID NO:2]) was generated from the above-mentioned wild-type protease (SEQ ID NO:1) in several rounds by random mutagenesis. Further rounds of random mutagenesis were set up on this mutant. The mutants C1-C7 were generated in these rounds of mutation. Therefore, all of the mutants mentioned here also carry the mutations of mutant C0C.

Variant	Amino acid substitutions relative to SEQ ID NO: 1					
C0C	P9T, N130D, T133A, N144K, N252T, Q271E					
C1	P9T, N130D, T133A, N144K, Y217M, N252T, Q271E	S189T	G166M			
C2	P9T, N130D, T133A, N144K, Y217M, N252T, Q271E	S189T	G166M	S224A		

20

-continued

Variant	Amino acid substitutions relative to SEQ ID NO: 1					
C3	P9T, N130D, T133A, N144K, Y217M, N252T, Q271E	S189T	S89A	G131H	S224A	
C5	P9T, N130D, T133A, N144K, Y217M, N252T, Q271E	S189T	Y6W	S224A		
C6	P9T, N130D, T133A, N144K, Y217M, N252T, Q271E	S211N	G166M			

Washing Agent Matrix Used

The following table specifies the washing agent matrix (commercially available, without enzymes, opt. brighteners, perfume and dyes) used for the washing test:

Chemical name	Wt. % of active substance in the raw material	Wt. % of active substance in the formulation
Demineralized water	100	Remainder
Alkyl benzene sulfonic acid	96	12-18
Anionic surfactants	70	4-8
C ₁₂ -C ₁₈ fatty acid Na salt	30	2-4
Non-ionic surfactants	100	8-14
Phosphonate	60	0.5-2
Citric acid	100	3-5
NaOH	50	0.5-2
Defoamer	100	<1%
Glycerol	99.5	1-3
1,2-propanediol	100	8-12
Monoethanolamine	100	4-8
Soil repellent polymer	30	0.5-1

Without opt. brighteners, perfume, dye and enzymes

Dosage 3.17 g/L

Protease Activity Assays

The activity of the protease is determined by the release of the chromophore para-nitroaniline from the substrate succinyl alanine-alanine-proline-phenylalanine-para-nitroanilide (AAPFPNA; Bachem L-1400). The release of the pNA causes an increase in absorbance at 410 nm, the temporal progression of which is a measure of the enzymatic activity.

The measurement was carried out at a temperature of 25° C., a pH of 8.6, and a wavelength of 410 nm. The measuring time was 5 minutes with a measuring interval of from 20 to 60 seconds.

Measurement Approach:

- 10 µL AAPF solution (70 mg/mL)
- 1000 µL Tris/HCl (0.1 M, pH 8.6 with 0.1% Brij 35)
- 10 µL diluted protease solution
- Kinetics created over 5 min at 25° C. (410 nm)

Storage Stability Test and Results

The proteases were stirred into a washing agent (see above) matrix at the same level of activity and stored at 30° C. By means of a conventional activity assay for proteases (hydrolysis of suc-AAPF-pNA), the starting activity and the residual activity of the protease are measured after 4 weeks' storage at 30° C. In order to generate harsh conditions, the proteases were stored in a washing agent matrix without a stabilizer.

The proteases were generated in shake flask supernatants from *Bacillus subtilis*. They were diluted to an equal level of activity. 50% washing agent matrix without boric acid was added to 50% of appropriately diluted *Bacillus subtilis* protease supernatant and mixed well. The sealed glasses were incubated at 30° C. At the time of sampling, a prede-

21

terminated amount of matrix/protease mixture was removed and dissolved by stirring for 20 min at RT in the sample buffer (0.1 M Tris/HCl, pH 8.6). The AAPF assay is then carried out as described above.

The claimed mutants have been found to be advantageous. The activity is shown in % of the residual activity of the starting variant (mutant 1 according to SEQ ID NO:2)—which already has significantly improved stability compared to the wild-type enzyme—after 4 weeks' storage at 30° C.

Variant	Residual activity
COC	100%
C1	179%

22

-continued

Variant	Residual activity
C2	173%
C3	115%
C5	176%
C6	139%

It can be seen that all the mutants, without the addition of boric acid, exhibit improved stability in comparison with the starting mutant according to SEQ ID NO:2. All the mutants exhibit a washing performance that is comparable to the wild type according to SEQ ID NO:1, i.e. they are at most 10% worse in terms of washing performance, which is within the measurement fluctuation (results not shown).

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 2

<210> SEQ ID NO 1

<211> LENGTH: 275

<212> TYPE: PRT

<213> ORGANISM: *Bacillus pumilus*

<400> SEQUENCE: 1

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Ala Gln Thr Val Pro Tyr Gly Ile Pro Gln Ile Lys Ala Pro Ala Val
1           5           10           15

His Ala Gln Gly Tyr Lys Gly Ala Asn Val Lys Val Ala Val Leu Asp
20          25          30

Thr Gly Ile His Ala Ala His Pro Asp Leu Asn Val Ala Gly Gly Ala
35          40          45

Ser Phe Val Pro Ser Glu Pro Asn Ala Thr Gln Asp Phe Gln Ser His
50          55          60

Gly Thr His Val Ala Gly Thr Ile Ala Ala Leu Asp Asn Thr Ile Gly
65          70          75          80

Val Leu Gly Val Ala Pro Ser Ala Ser Leu Tyr Ala Val Lys Val Leu
85          90          95

Asp Arg Asn Gly Asp Gly Gln Tyr Ser Trp Ile Ile Ser Gly Ile Glu
100         105         110

Trp Ala Val Ala Asn Asn Met Asp Val Ile Asn Met Ser Leu Gly Gly
115         120         125

Pro Asn Gly Ser Thr Ala Leu Lys Asn Ala Val Asp Thr Ala Asn Asn
130         135         140

Arg Gly Val Val Val Val Ala Ala Ala Gly Asn Ser Gly Ser Thr Gly
145         150         155         160

Ser Thr Ser Thr Val Gly Tyr Pro Ala Lys Tyr Asp Ser Thr Ile Ala
165         170         175

Val Ala Asn Val Asn Ser Ser Asn Val Arg Asn Ser Ser Ser Ser Ala
180         185         190

Gly Pro Glu Leu Asp Val Ser Ala Pro Gly Thr Ser Ile Leu Ser Thr
195         200         205

Val Pro Ser Ser Gly Tyr Thr Ser Tyr Thr Gly Thr Ser Met Ala Ser
210         215         220

Pro His Val Ala Gly Ala Ala Ala Leu Ile Leu Ser Lys Asn Pro Asn
225         230         235         240

Leu Ser Asn Ser Gln Val Arg Gln Arg Leu Glu Asn Thr Ala Thr Pro
245         250         255

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-continued

Leu Gly Asn Ser Phe Tyr Tyr Gly Lys Gly Leu Ile Asn Ala Gln Ala
 260 265 270

Ala Ser Asn
 275

<210> SEQ ID NO 2
 <211> LENGTH: 275
 <212> TYPE: PRT
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: Protease variant C0C

<400> SEQUENCE: 2

Ala Gln Thr Val Pro Tyr Gly Ile Thr Gln Ile Lys Ala Pro Ala Val
 1 5 10 15

His Ala Gln Gly Tyr Lys Gly Ala Asn Val Lys Val Ala Val Leu Asp
 20 25 30

Thr Gly Ile His Ala Ala His Pro Asp Leu Asn Val Ala Gly Gly Ala
 35 40 45

Ser Phe Val Pro Ser Glu Pro Asn Ala Thr Gln Asp Phe Gln Ser His
 50 55 60

Gly Thr His Val Ala Gly Thr Ile Ala Ala Leu Asp Asn Thr Ile Gly
 65 70 75 80

Val Leu Gly Val Ala Pro Ser Ala Ser Leu Tyr Ala Val Lys Val Leu
 85 90 95

Asp Arg Asn Gly Asp Gly Gln Tyr Ser Trp Ile Ile Ser Gly Ile Glu
 100 105 110

Trp Ala Val Ala Asn Asn Met Asp Val Ile Asn Met Ser Leu Gly Gly
 115 120 125

Pro Asp Gly Ser Ala Ala Leu Lys Asn Ala Val Asp Thr Ala Asn Lys
 130 135 140

Arg Gly Val Val Val Val Ala Ala Ala Gly Asn Ser Gly Ser Thr Gly
 145 150 155 160

Ser Thr Ser Thr Val Gly Tyr Pro Ala Lys Tyr Asp Ser Thr Ile Ala
 165 170 175

Val Ala Asn Val Asn Ser Ser Asn Val Arg Asn Ser Ser Ser Ser Ala
 180 185 190

Gly Pro Glu Leu Asp Val Ser Ala Pro Gly Thr Ser Ile Leu Ser Thr
 195 200 205

Val Pro Ser Ser Gly Tyr Thr Ser Tyr Thr Gly Thr Ser Met Ala Ser
 210 215 220

Pro His Val Ala Gly Ala Ala Ala Leu Ile Leu Ser Lys Asn Pro Asn
 225 230 235 240

Leu Ser Asn Ser Gln Val Arg Gln Arg Leu Glu Thr Thr Ala Thr Pro
 245 250 255

Leu Gly Asn Ser Phe Tyr Tyr Gly Lys Gly Leu Ile Asn Ala Glu Ala
 260 265 270

Ala Ser Asn
 275

25

The invention claimed is:

1. A protease comprising:

a conservatively mutated amino acid sequence having at least 70% sequence identity with the amino acid sequence given in SEQ ID NO:1 over its entire length; wherein the protease has, in each case based on the numbering according to SEQ ID NO:1,

(a) amino acid substitutions at the positions corresponding to positions 9, 130, 133, 144, 217, 252, and 271; as well as

(b) one or more further amino acid substitutions at the positions corresponding to positions 6, 89, 131, 166, 189, 211, and 224; and,

wherein the protease exhibits improved washing performance when compared to the protease of SEQ ID NO:1.

2. The protease according to claim 1, wherein

(1) the amino acid substitutions occur at 9T, 130D/V, 133A, 144K, 217M, 252T, and 271E, at the positions corresponding to positions 9, 130, 133, 144, 217, 252, and 271; and

(2) the one or more further amino acid substitutions are selected from 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q, and 224A/G at the positions corresponding to positions 6, 89, 131, 166, 189, 211, and 224.

3. The protease according to claim 1, wherein the

(1) amino acid substitutions occur at 9T, 130D/V, 133A, 144K, 217M, 252T and 271E for the positions corresponding to positions 9, 130, 133, 144, 217, 252, and 271; and

(2) one or more further amino acid substitutions occur at (a) 166M and 189T for the positions corresponding to positions 166 and 189;

(b) 166M, 189T, and 224A for the positions corresponding to positions 166, 189, and 224;

(c) 89A, 131H, 189T and 224A for the positions corresponding to positions 89, 131, 189, and 224;

(d) 89A, 189T, and 224A for the positions corresponding to positions 89, 189, and 224;

(e) 6W, 189T, and 224A for the positions corresponding to positions 6, 189, and 224;

(f) 166M and 211N for the positions corresponding to positions 166 and 211;

(g) 6W and 189T for the positions corresponding to positions 6 and 189;

(h) 131H and 189T for the positions corresponding to positions 131 and 189;

(i) 89A and 131H for the positions corresponding to positions 89 and 131;

(j) 89A and 166M for the positions corresponding to positions 89 and 166;

(k) 89A and 189T for the positions corresponding to positions 89 and 189;

(l) 89A and 211N for the positions corresponding to positions 89 and 211;

(m) 89A and 224A for the positions corresponding to positions 89 and 224;

(n) 131H and 166M for the positions corresponding to positions 131 and 166;

(o) 131H and 211N for the positions corresponding to positions 131 and 211;

26

(p) 131H and 224A for the positions corresponding to positions 131 and 224; or

(g) 189T and 224 for the positions corresponding to positions 189 and 224.

4. The protease according to claim 3, wherein the amino acid substitutions occur at 9T, 130D, 133A, 144K, 217M, 252T and 271E for the positions corresponding to positions 9, 130, 133, 144, 217, 252 and 271; and wherein the one or more further amino acid substitutions occur at

(a) 166M and 189T for the positions corresponding to positions 166 and 189;

(b) 166M, 189T and 224A for the positions corresponding to positions 166, 189, and 224;

(c) 89A, 131H and 189T for the positions corresponding to positions 89, 131, and 189;

(d) 89A and 189T for the positions corresponding to positions 89 and 189;

(e) 6W, 189T and 224A for the positions corresponding to positions 6, 189, and 224; or

(f) 166M and 211N for the positions corresponding to positions 166 and 211.

5. The protease according to claim 3, wherein the amino acid substitutions 9T, 130V, 133A, 144K, 217M, 252T, and 271E for the positions corresponding to positions 9, 130, 133, 144, 217, 252, and 271; and wherein the one or more further amino acid substitutions occur at 89A, 131H, and 189T at the positions corresponding to positions 89, 131, and 189.

6. The protease according to claim 1, further comprising a one or more conservative amino acid substitutions, fragmentation, deletion, insertion mutagenesis, substitution mutagenesis, and combinations thereof.

7. A nucleic acid that codes for a protease according to claim 1.

8. A non-human host cell containing a protease according to claim 1.

9. A method for producing a protease, wherein the method comprises:

a) cultivating a host cell according to claim 8; and

b) isolating the protease from the culture medium or from the host cell.

10. A washing or cleaning agent comprising at least one protease according to claim 1.

11. A method for cleaning textiles or hard surfaces, wherein the method comprises applying an agent according to claim 10.

12. The protease according to claim 1, wherein the amino acid substitutions are 9T, 130D/V, 133A, 144K, 217M, 252T, and 271E corresponding to the amino acid substitutions at the positions corresponding to positions 9, 130, 133, 144, 217, 252, and 272, respectively.

13. The protease according to claim 1, wherein the one or more further amino acid substitutions are selected from 6W/F, 89A/G, 131H/Y/F, 166M/L/I, 189T/L/I, 211N/Q, and 224A/G corresponding to positions 6, 89, 131, 166, 189, 211, and 224, respectively.

14. The protease according to claim 1, wherein the one or more further amino acid substitutions are selected from 6W, 89A, 131H, 166M, 189T, 211N, and 224A corresponding to positions 6, 89, 131, 166, 189, 211, and 224, respectively.

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